ADITYA UNIVERSITY

SCHOOL OF ENGINEERING

Department of Computer Science & Engineering

CERTIFICATE

This is to certify that the Cornerstone project work entitled *"***ONLINE SHOPPING CART SIMULATION***"* is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Computer Science and Engineering for the academic year 2025-2026.

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DECLARATION

We hereby declare that the Cornerstone project entitled "**ONLINE SHOPPING CART SIMULATION**" is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree** in **Computer Science & Engineering**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

Place: Surampalem

Date:

By

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ABSTRACT

The Online Shopping Cart Simulation (ShopVibe) is a desktop-based e-commerce system developed to provide an efficient and user-friendly platform for managing shopping activities. The system allows users to register, log in, browse products, add items to a cart, and place orders seamlessly. The application also provides administrative functionalities such as managing products, viewing orders, and monitoring registered users through a dedicated user management panel. The system also provides account management features where users can update their personal details and manage their accounts.

The application is developed using Java with Swing for the graphical user interface, MySQL for database management, and JDBC for connectivity. Additionally, iText PDF library is used for generating bills in PDF format after checkout.

The system ensures data accuracy, reduces manual effort, and enhances the overall shopping experience by providing features like search, category filtering, cart management, and order history. Security is maintained using authentication and prepared statements to prevent SQL injection.

Overall, the Online Shopping Cart Simulation(ShopVibe) serves as a reliable and scalable solution for basic e-commerce operations and demonstrates the practical implementation of database-driven applications using Java.

LIST OF FIGURES

LIST OF TABLES

ACRONYMS

Acronym	Full Form
GUI	Graphical User Interface
JDBC	Java Database Connectivity
SQL	Structured Query Language
DBMS	Database Management System
PDF	Portable Document Format

CHAPTER 1 INTRODUCTION

1.1 Brief Information about the Project

The Online Shopping Cart Simulation (ShopVibe) is a desktop-based e-commerce system developed to simplify and digitize the process of online shopping. The application provides a platform where users can register, log in, browse products, add items to a cart, and place orders efficiently.

The system supports two types of users: Customers and Admin. Customers can explore products by category, search for items, manage their shopping cart, and complete purchases with bill generation. The Admin has control over product management, including adding, updating, and deleting products, as well as viewing customer orders and registered customers.

The application is developed using Java for core programming, Swing for graphical user interface design, MySQL for database management, and JDBC for connectivity. It also integrates iText PDF library to generate bills after checkout. This project replaces manual shopping processes with an automated system, improving efficiency, accuracy, and user experience.

1.2 Motivation and Contribution of the Project

In today's digital era, online shopping has become a necessity. However, many small-scale systems lack proper management of products, user data, and order processing. Manual methods or poorly designed systems often lead to issues such as data inconsistency, difficulty in tracking orders, and poor user experience.

The motivation behind developing the Shopping Cart Application is to create a simple, efficient, and user-friendly e-commerce solution that demonstrates real-world application development. The key contributions of this project include:

- Providing a complete e-commerce workflow from login to checkout
- Reducing manual effort in managing products and orders
- Ensuring data consistency using a relational database
- Improving user experience through an interactive GUI
- Implementing secure authentication and role-based access
- Generating automated PDF bills for transactions

This project helps in understanding how real-world shopping systems function and provides practical exposure to database-driven application development.

1.3 Objectives of the Project

The main objectives of the Shopping Cart Application are:

- To develop a user-friendly application for online shopping
- To implement secure user authentication and authorization
- To allow users to browse and search products easily
- To provide shopping cart functionality with quantity management
- To enable order placement and bill generation
- To allow admin users to manage products and orders efficiently
- To store and manage data using a structured database

These objectives ensure that the system performs all essential e-commerce operations effectively.

1.4 Scope of the Project

The scope of the Online Shopping Cart Simulation (ShopVibe) includes providing a basic yet complete e-commerce solution for users and administrators. The system can be used in small-scale environments such as local businesses, college projects, or learning platforms. It supports functionalities like user registration, login, product browsing, cart management, order placement, and order history tracking.

For administrators, the system provides features to manage products and monitor customer orders. The application ensures smooth interaction between users and the database through a graphical interface.

However, the system is limited to a desktop environment and does not include advanced features like online payment gateways or real-time delivery tracking. These can be added in future enhancements. Overall, the project demonstrates the design and implementation of a database-driven application with real-world relevance.

CHAPTER 2 SYSTEM ANALYSIS

2.1 Functional Requirements

Functional requirements define the core functionalities that the Online Shopping Cart Simulation(ShopVibe) must perform.

User Registration

The system should allow new users to register by providing details such as username, email, phone number, and address. Proper validation must be applied to ensure correct data entry.

User Login

The system should authenticate users using their username and password. Based on the role (Customer / Admin), appropriate access should be provided.

User Settings

Users should be able to update personal details (email, phone, address, password) and delete their account.

Product Browsing

Users should be able to view products categorized into different sections such as Electronics, Clothing, Books, etc.

Product Search

The system should provide a search feature to find products by name quickly and efficiently.

Shopping Cart Management

Users should be able to add products to the cart, update product quantity, remove items from the cart, and view the total price. The system should prevent duplicate entries by updating quantity instead.

Checkout Process

Users should be able to place orders and complete the checkout process. After checkout, a PDF bill should be generated automatically.

Order History

Users should be able to view their previous orders and re-download bills anytime.

Admin Product Management

Admin should be able to add new products, update existing product details, and delete products.

Admin Order Management

Admin should be able to view all customer orders, view order items, and export order details to a file.

Admin User Panel

Admin should be able to view all registered users in the system.

2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

Performance

The system should respond quickly to user actions such as login, search, and checkout.

Security

The system should ensure secure authentication and protect data using techniques like prepared statements to prevent SQL injection.

Usability

The application should have a simple and user-friendly interface so that users can easily navigate and perform operations.

Reliability

The system should function correctly without crashes and handle errors properly.

Scalability

The system should be able to handle increasing users and product data without performance degradation.

Maintainability

The system should be easy to update and modify in the future.

2.3 Requirements Specification

This section describes the hardware and software requirements needed to run the application.

2.3.1 Minimum Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB minimum
- Hard Disk: 20 GB free space
- Monitor: Standard display
- Input Devices: Keyboard and Mouse

2.3.2 Software Requirements

- Operating System: Windows 10/11
- Programming Language: Java (JDK 11 or higher)
- GUI Framework: Java Swing
- Database: MySQL 8.0 or above
- Connectivity: JDBC
- PDF Library: iText PDF
- Build Tool: Apache Ant
- IDE: NetBeans

CHAPTER 3 TECHNOLOGY DESCRIPTION

3.1 Programming Language

3.1.1 Introduction to Java

Java is a high-level, object-oriented programming language developed by Sun Microsystems. It is widely used for building desktop, web, and enterprise applications due to its platform independence and robustness.

In the Online Shopping Cart Simulation (ShopVibe), Java is used as the core programming language to implement the application logic. It handles operations such as user authentication, product management, cart operations, and order processing. Java provides important features such as:

- Object-Oriented Programming (OOP)
- Exception Handling
- Platform Independence (Write Once, Run Anywhere)
- Strong Memory Management

These features make Java suitable for developing reliable and scalable applications like ShopVibe.

3.2 Framework / Libraries

3.2.1 Introduction to Swing

Swing is a part of Java used to build Graphical User Interfaces (GUI). It provides a wide range of components such as buttons, labels, text fields, tables, and panels. In this project, Swing is used to design interactive user interfaces such as:

- Login and Signup Screens
- Product Catalog
- Shopping Cart
- Admin Dashboard
- Order History

Swing helps in creating a visually interactive and user-friendly application. It also allows customisation of layouts, colours, and components to enhance the user experience.

3.2.2 iText PDF Library

iText is a Java library used for generating and manipulating PDF documents. In ShopVibe, iText is used to generate bills after checkout, format transaction details professionally, and automatically open or download the bill. This feature enhances the practicality of the application by providing users with a proper invoice for their purchases.

3.3 Database Technology

The ShopVibe application uses a relational database to store and manage data such as users, products, cart items, and orders. The database ensures structured storage, easy retrieval, and data consistency.

3.3.1 JDBC (Java Database Connectivity)

JDBC is an API used to connect Java applications with databases. It acts as a bridge between the application and the database. In this project, JDBC is used to:

- Establish database connections
- Execute SQL queries (INSERT, UPDATE, DELETE, SELECT)
- Retrieve and process data
- Close database connections properly

Prepared statements are used to improve security and prevent SQL injection attacks.

3.3.2 MySQL Database

MySQL is an open-source relational database management system widely used for storing structured data. In ShopVibe, MySQL is used to store user information, product details, cart data, orders, and order items. MySQL provides:

- Fast data access
- Data integrity using constraints
- Scalability for handling large data

It ensures that all application data is stored securely and efficiently.

3.4 Tools & IDEs

3.4.1 Development Tools

The following tools are used for developing the ShopVibe application:

- **NetBeans** – Used for writing and managing Java code.
- **MySQL Workbench / XAMPP Control Panel** – Used for designing and managing the database.
- **Apache Ant –** Used for building and running the project.

These tools help in improving productivity, debugging, and efficient project management.

CHAPTER 4 DATABASE DESIGN

4.1 Database Tables

The Online Shopping Cart Simulation (ShopVibe) uses a relational database to store and manage data efficiently. The database consists of the following main tables:

- **Users Table** – Stores user details such as login credentials and personal information. Used in the Admin User Panel and Customer Settings Panel.
- **Products Table** – Stores product details like name, price, category, and stock.
- **Cart Table** – Stores items added to the user's cart.
- **Orders Table** – Stores order details after checkout.
- **Order_Items Table** – Stores individual items for each order.

This structured design helps in reducing redundancy and maintaining data consistency.

4.2 Table Structure

Users Table

Field Name	Data Type	Description	Constraint
id	INT	User ID	Primary Key
username	VARCHAR(100)	Username	Unique, Not Null
password	VARCHAR(100)	Password	Not Null
email	VARCHAR(100)	Email Address	Unique, Not Null
phone	VARCHAR(20)	Phone Number	Not Null
address	TEXT	User Address	Not Null
role	VARCHAR(20)	User Role	Default 'customer'

Products Table

Field Name	Data Type	Description	Constraint
id	INT	Product ID	Primary Key
name	VARCHAR(150)	Product Name	Not Null
description	TEXT	Product Description	—
price	DECIMAL(10,2)	Product Price	Not Null

Field Name	Data Type	Description	Constraint
stock	INT	Available Quantity	Not Null
category	VARCHAR(50)	Product Category	—

Cart Table

Field Name	Data Type	Description	Constraint
id	INT	Cart ID	Primary Key
user_id	INT	User ID	Foreign Key
product_id	INT	Product ID	Foreign Key
quantity	INT	Quantity	Not Null

Orders Table

Field Name	Data Type	Description	Constraint
id	INT	Order ID	Primary Key
user_id	INT	User ID	Foreign Key
order_date	TIMESTAMP	Order Date	Default Current
total_amount	DECIMAL(10,2)	Total Price	Not Null
status	VARCHAR(50)	Order Status	Default 'completed'

Order Items Table

Field Name	Data Type	Description	Constraint
id	INT	Item ID	Primary Key
order_id	INT	Order ID	Foreign Key
product_id	INT	Product ID	Foreign Key
quantity	INT	Quantity	Not Null
price	DECIMAL(10,2)	Price	Not Null

4.3 Keys and Constraints

Keys and constraints are used to maintain data integrity and consistency in the database.

Primary Keys

- id in Users Table
- id in Products Table
- id in Cart Table
- id in Orders Table
- id in Order_Items Table

Foreign Keys

- user_id in Cart and Orders tables references Users(id)
- product_id in Cart and Order_Items tables references Products(id)
- order_id in Order_Items references Orders(id)

Constraints

- NOT NULL: Ensures required fields are not empty
- UNIQUE: Prevents duplicate usernames and emails
- DEFAULT: Provides default values (e.g., role, status)

These constraints ensure accurate and valid data storage.

4.4 Table Relationships

The tables in the ShopVibe database are interconnected as follows:

- One User can have multiple Orders (One-to-Many)
- One Order can contain multiple Order Items (One-to-Many)
- One Product can appear in multiple Cart and Order Items (One-to-Many)
- Each Cart Item belongs to one user and one product

This relational structure ensures efficient data organisation and supports complex operations like order tracking and cart management.

CHAPTER 5 SYSTEM DESIGN

5.1 System Architecture

The Online Shopping Cart Simulation (ShopVibe) follows a three-layer architecture to ensure proper organization, maintainability, and scalability of the system.

Presentation Layer (User Interface)

This layer is responsible for user interaction and is developed using Java Swing. It includes all GUI components such as Login and Signup screens, Product Catalog, Shopping Cart interface, Admin Dashboard, and Order History. This layer collects user input and displays the output in a user-friendly format.

Business Logic Layer

This layer contains the core logic of the application. It processes user inputs and performs operations such as user authentication, product search and filtering, cart management, order processing, and admin operations. It acts as a bridge between the user interface and the database.

Data Layer (Database)

This layer is responsible for storing and managing data using MySQL database. It includes tables such as users, products, cart, orders, and order_items. The application communicates with this layer using JDBC to perform database operations like inserting, updating, deleting, and retrieving data.

This layered architecture improves:

- Code organization
- Security
- Ease of maintenance
- Scalability of the application

5.2 Modules Description

The ShopVibe system is divided into several modules for better organization and functionality.

User Management Module

- Handles user registration and login
- Validates user input
- Manages authentication and roles (Customer / Admin)

Product Management Module

- Allows admin to add, update, and delete products
- Displays product details to users
- Organizes products by categories

Customer Settings Module

- Allows users to manage their personal information
- Users can update email, phone number, address, and password
- Provides an option to delete the account securely

Shopping Cart Module

- Allows users to add products to the cart
- Updates quantity of items
- Removes items from the cart
- Calculates total price

Order Management Module

- Handles checkout process
- Stores order details in database
- Generates PDF bills
- Maintains order history

Admin Module

- Provides access to admin dashboard
- Allows viewing all customer orders
- Enables export of order data
- Allows the admin to view all registered users

Database Connectivity Module

- Manages connection between Java application and MySQL database
- Executes SQL queries using JDBC
- Ensures proper closing of connections

User Interface Module

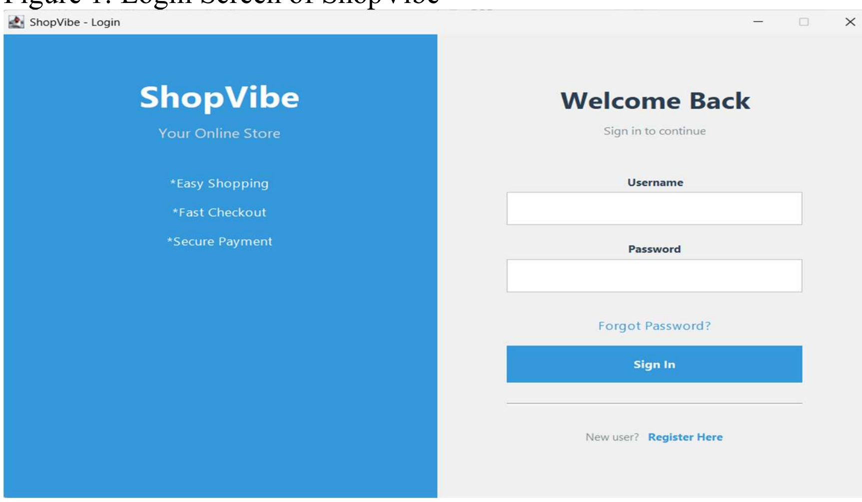
- Designs all application screens using Swing
- Handles user interaction
- Displays messages and feedback

CHAPTER 6 SYSTEM IMPLEMENTATION

Login Screen

The Login Screen allows users to enter their username and password to access the system. Based on the credentials, the system authenticates the user and redirects them to either the Customer Dashboard or Admin Dashboard. If users forget password they can reset their password (validation done through user name and email)

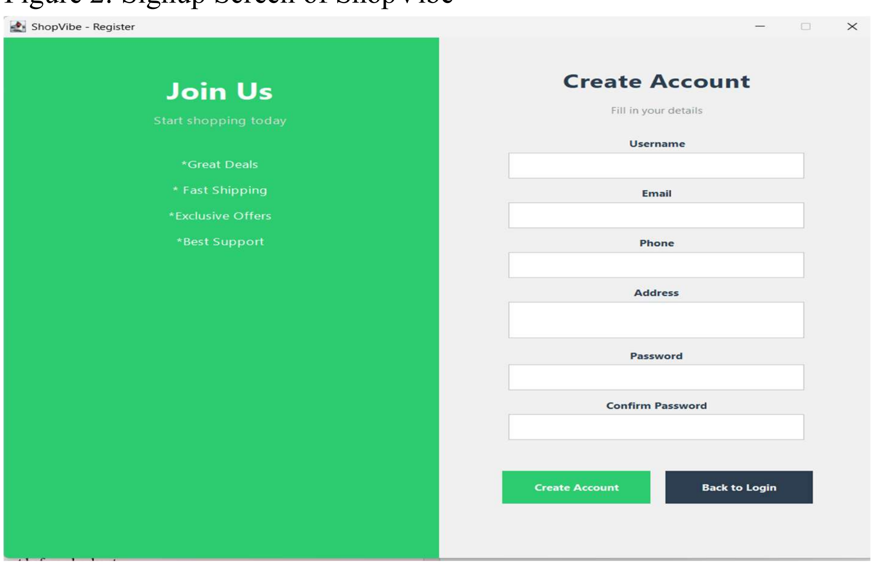
Figure 1: Login Screen of ShopVibe



Signup Screen

The Signup Screen allows new users to register by entering required details such as username, email, phone number, and address. Validation is applied to ensure correct data entry.

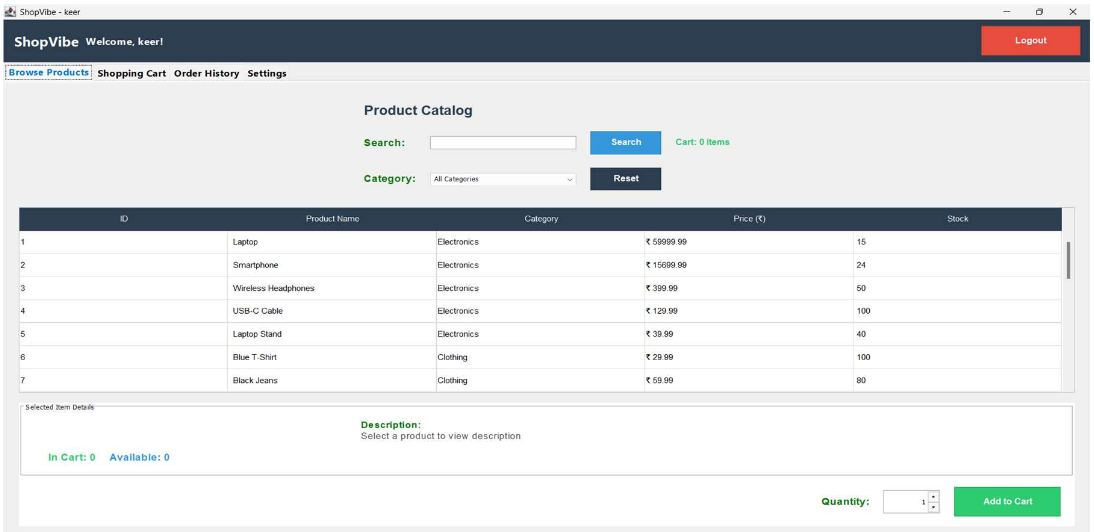
Figure 2: Signup Screen of ShopVibe



Product Catalog

This screen displays all available products categorised into different sections. Users can search for products and view details such as price and description.

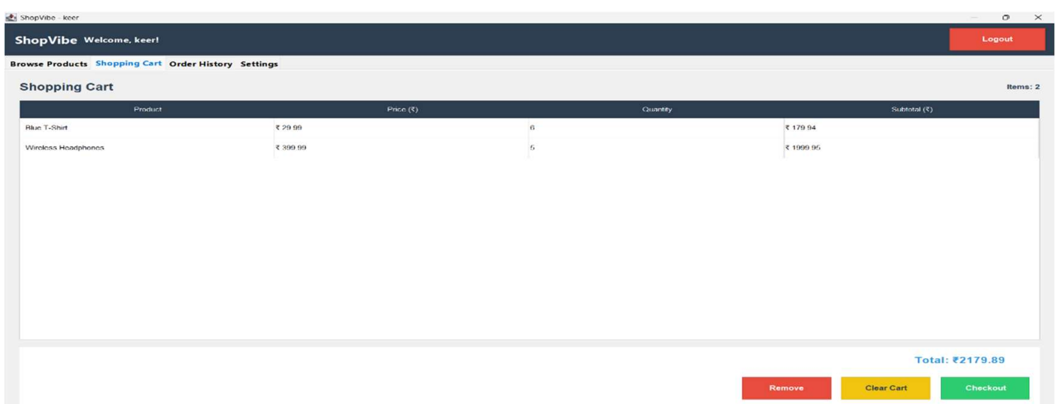
Figure 3: Product Catalog of ShopVibe



Shopping Cart

The Shopping Cart screen displays all items added by the user. Users can update quantities, remove items, and view the total cost before checkout.

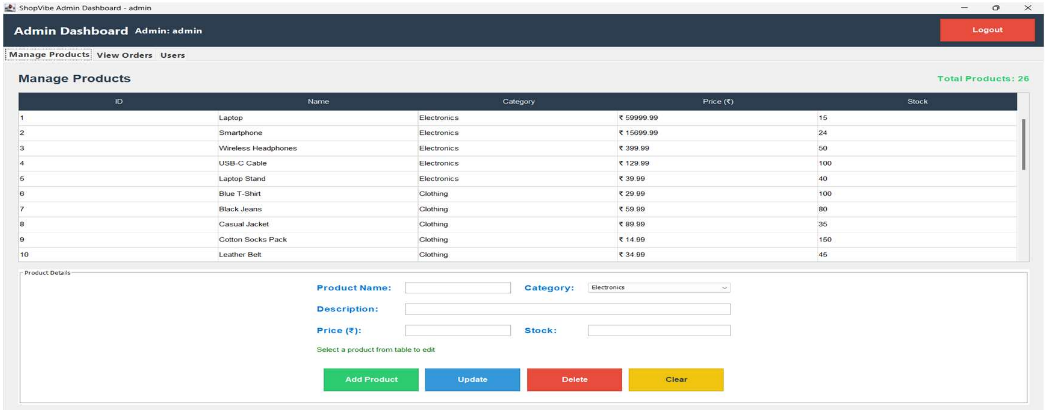
Figure 4: Shopping Cart of ShopVibe

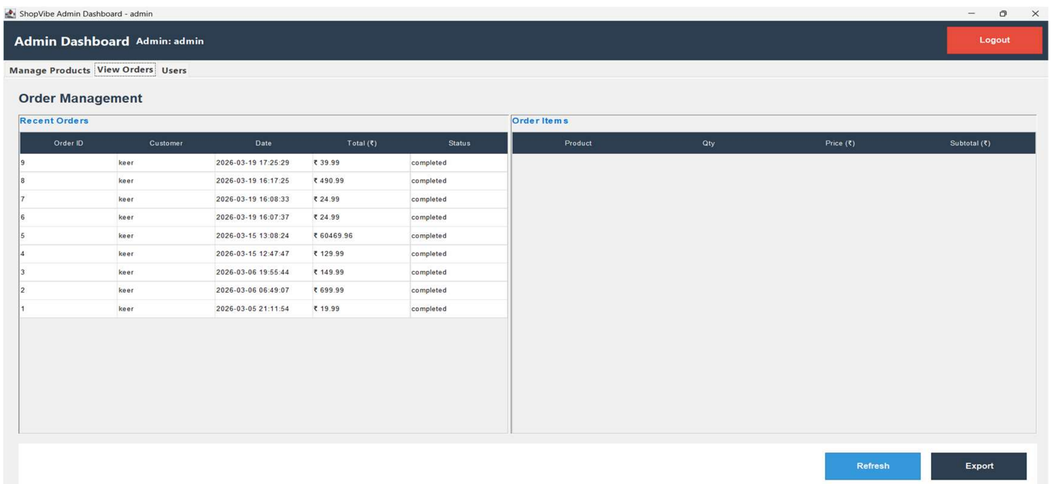


Admin Dashboard

The Admin Dashboard provides functionalities for managing products and viewing customer orders. Admin can add, update, or delete products and monitor system activity.

Figure 5: Admin Dashboard of ShopVibe

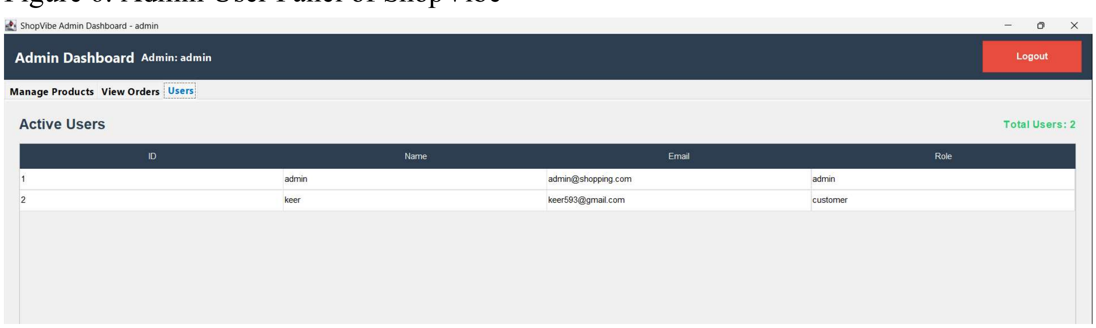




Admin User Panel

The Admin User Panel is designed to display all registered users in the system. It uses a JTable to present user details such as ID, username, email, and role in a structured format. The panel also shows the total number of registered users. The data is fetched from the database via JDBC and displayed dynamically.

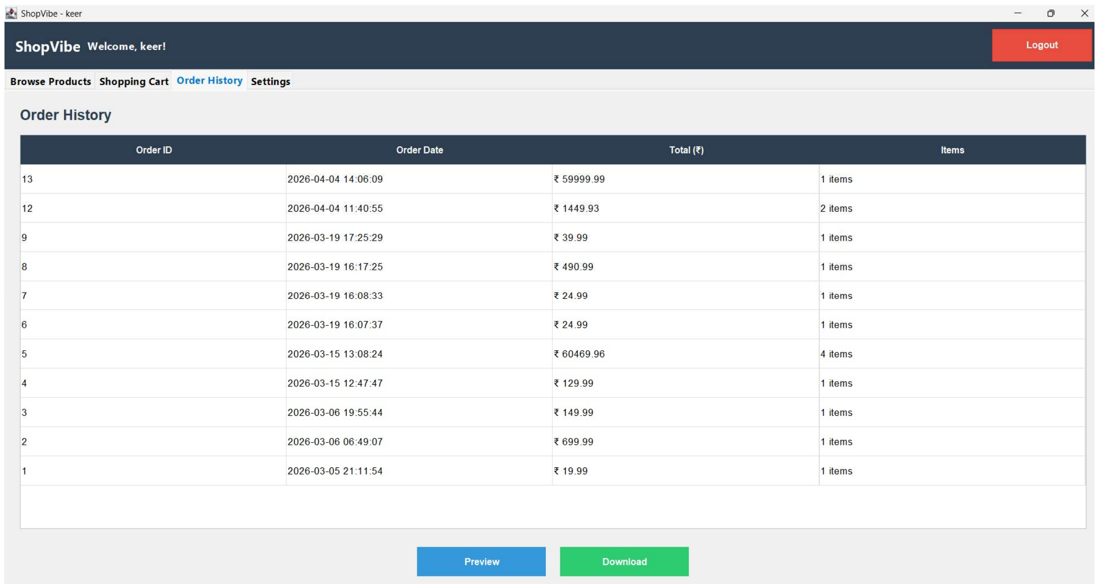
Figure 6: Admin User Panel of ShopVibe



Order History

This screen allows users to view their previous orders , preview and re-download PDF bills anytime.

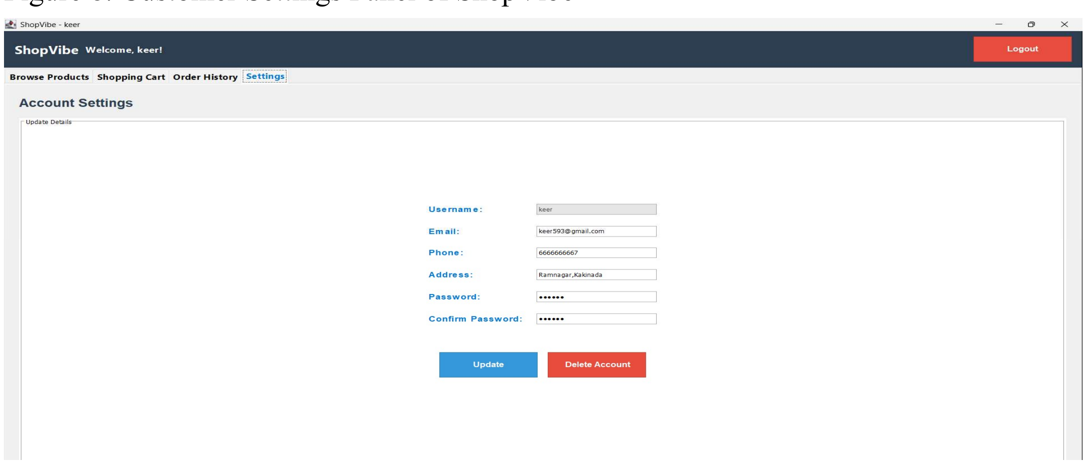
Figure 7: Order History of ShopVibe



Customer Settings Panel

The Customer Settings Panel allows users to view and update their account details. It includes fields such as email, phone number, address, and password. The system performs validation checks to ensure correct data entry. Users can update their details, and the changes are saved in the database using JDBC. The panel also provides an option to delete the user account after confirmation. Upon deletion, all related data is removed from the system.

Figure 8: Customer Settings Panel of ShopVibe



CHAPTER 7 CONCLUSION

The Online Shopping Cart Simulation (ShopVibe) has been successfully developed to provide an efficient and user-friendly platform for online shopping. The system replaces traditional manual methods with a digital solution, thereby improving accuracy, reducing effort, and enhancing user experience.

The application provides essential features such as user authentication, product browsing, cart management, order processing, and PDF bill generation. It also includes an admin module for managing products and viewing orders.

By using technologies like Java, Swing, JDBC, MySQL, and iText PDF, the system ensures reliable performance, secure data handling, and efficient database management.

Overall, the project meets its objectives and demonstrates the practical implementation of a real-world e-commerce system. It also provides a strong foundation for further enhancements and scalability.

CHAPTER 8 FUTURE ENHANCEMENTS

The Online Shopping Cart Simulation (ShopVibe) can be further improved by adding advanced features to enhance functionality and user experience. Some possible future enhancements include:

- Integration of online payment gateways (UPI, Cards, Net Banking)
- Email notifications for order confirmation
- Product reviews and ratings system
- Wishlist feature for users
- Discount coupons and offers
- Inventory management system
- Multiple delivery addresses
- Order tracking system
- Admin analytics dashboard

These enhancements can make the system more powerful, scalable, and suitable for real-world deployment.

CHAPTER 9 REFERENCES

The following resources were used during the development of the project:

[1] Herbert Schildt, *Java: The Complete Reference*, Oracle Press.

[2] E. Balagurusamy, *Programming with Java*, McGraw Hill.

[3] Korth, Silberschatz, Sudarshan, *Database System Concepts*.

[4] Oracle Java Documentation: https://docs.oracle.com/javase

[5] MySQL Documentation: https://dev.mysql.com/doc

[6] JDBC Tutorial – GeeksforGeeks.

[7] Java Swing Tutorial – TutorialsPoint.

[8] iText PDF Documentation.

APPENDIX

A.1 Sample Code

Database Connection Code

```
package database;
import java.sql.Connection;
import java.sql.DriverManager;

public class DatabaseConnection {
    public static Connection getConnection() {
        Connection con = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/shopping_cart_db",
                "root", "");
            System.out.println("Database connected successfully!");
        } catch (Exception e) {
            System.out.println("Database connection failed!");
            e.printStackTrace();
        }
        return con;
    }
}
```

Insert Product Example

```
String query = "INSERT INTO products(name, price, stock) VALUES (?, ?, ?)";
PreparedStatement pst = con.prepareStatement(query);
pst.setString(1, "Laptop");
pst.setDouble(2, 50000);
pst.setInt(3, 10);
pst.executeUpdate();
```

A.2 SQL Queries

Create Products Table

```
CREATE TABLE products (
    id       INT PRIMARY KEY AUTO_INCREMENT,
    name     VARCHAR(150)   NOT NULL,
    price    DECIMAL(10,2)  NOT NULL,
    stock    INT            NOT NULL,
    category VARCHAR(50)
);
```

Create Users Table

```
CREATE TABLE users (
    id       INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(100) UNIQUE NOT NULL,
    password VARCHAR(100)        NOT NULL,
    email    VARCHAR(100) UNIQUE NOT NULL,
    phone    VARCHAR(20)         NOT NULL,
    address  TEXT                NOT NULL,
    role     VARCHAR(20) DEFAULT 'customer'
);
```